

T HARISH REDDY

Santhipuram

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PROFILE SUMMARY

Trainee Data Engineer with hands-on experience in designing and optimizing data workflows using Python and PySpark. Skilled in managing medium to large datasets in distributed environments, with practical exposure to ETL pipeline development and data transformation. Have a basic understanding of implementing machine learning models using Spark MLlib, with minimal experience in data visualization using Matplotlib and supporting database operations with MySQL.

EDUCATION

Gandhi Institute Of Technology And Management **Aug 2022 – Present**
B Tech, CSE AIML - CGPA - 7.35 *Bengaluru*

Mother Teresa Junior College **June 2020 – May 2022**
Board of Intermediate Education - Percentage - 71.1% *Palamaner*

Vijaya Vani (EM) High School **June 2019 – Mar 2020**
Board of Secondary Education - Percentage - 80.16% *Chowdepalli*

PROJECTS

Classification of Normal/Cancer Class Lung CT

- Designed and implemented a CNN-based model to classify lung CT images into normal and cancerous classes, utilizing feature extraction and end-to-end learning for accurate prediction.
- Developed deep learning-based classification of lung CT images using CNN architectures such as VGG16, ResNet50, and DenseNet, enabling reliable detection of cancerous and normal cases.
- Validated model performance through cross-validation and confusion matrix metrics, where **DenseNet** achieved the best results with **98.75% accuracy**, **97.56% precision**, **100% recall (sensitivity)**, and a **98.7% F1-score** on the benchmark dataset.
- Tools Used:** Python
- Algorithms Used:** VGG16, ResNet50, DenseNet

AI-Powered Supply Chain Control Tower – Real-Time Disruption Prediction & Dynamic Routing

- Developed an end-to-end AI-powered supply chain control tower to proactively detect and mitigate logistics disruptions, enabling early risk prediction 24–48 hours in advance.
- Engineered scalable ETL pipelines to process 10M+ records from heterogeneous sources (GPS/IoT, weather, traffic, port congestion), ensuring reliable data ingestion, validation, and storage in PostgreSQL.
- Implemented machine learning ensemble models (Random Forest, Gradient Boosting) to predict high-risk shipments, along with Ridge Regression for time-series ETA forecasting.
- Designed a dynamic route optimization engine using Dijkstra and A* algorithms integrated with a Reinforcement Learning-based policy for cost, time, and risk-aware routing.
- Built a real-time streaming system using WebSockets and Flask to deliver live shipment tracking, risk heatmaps, KPIs, and alerts via an interactive dashboard.
- Containerized and deployed the system using Docker and Kubernetes, ensuring scalability, fault tolerance, and production readiness.
- Technologies:** Python, Machine Learning, Random Forest, Gradient Boosting, Ridge Regression, Reinforcement Learning (Basics), Flask, WebSockets, PostgreSQL, Docker, Kubernetes

Urban Heat Island Analysis – Bengaluru

- Applied Google Earth Engine and ArcMap to preprocess, analyze, and visualize large-scale spatial data for accurate mapping of Bengaluru's Urban Heat Island (UHI) effect.
- Built a Random Forest Regression model integrating environmental factors to predict urban heat patterns, identifying hotspots for climate adaptation and sustainable city planning.
- Evaluated model performance using statistical metrics, achieving **95% accuracy**, an **R² score of 0.94**, **RMSE of 1.8**, and **MAE of 1.2**, confirming the robustness and reliability of the prediction model.
- Algorithms Used:** Random Forest Regression

INTERNSHIP

Andor Tech

Dec 2025 – Present

Python Developer

Bengaluru, India

- Developed AI-driven systems leveraging Large Language Models (LLMs) with agent-based architectures to automate complex workflows and intelligent decision-making tasks.
- Implemented Retrieval-Augmented Generation (RAG) pipelines for semantic search and context-aware question answering over structured and unstructured datasets.
- Built document processing pipelines using OCR tools such as Tesseract to extract, clean, and structure data from PDFs and scanned documents.
- Worked with Vision-Language Models (VLMs) to process and analyze multimodal inputs, enabling combined text and image understanding for enhanced automation.
- Designed embedding pipelines and similarity search systems for efficient information retrieval using vector representations.
- Performed data preprocessing, transformation, and validation to ensure high-quality inputs for machine learning and LLM-based applications.
- Collaborated with teams to improve system scalability, accuracy, and performance across AI-driven data pipelines.
- **Skills:** Python, LLMs, RAG, AI Agents, VLMs, OCR (Tesseract), Embeddings, Vector Databases, Data Processing

Andor Tech

May 2025 – Oct 2025

Trainee Data Engineer

Bengaluru, India

- Developed and optimized data processing workflows using Python, Pandas, and PySpark to efficiently handle large-scale datasets in distributed environments, manipulating and transforming data across multiple formats to ensure quality and consistency.
- Designed and implemented ETL pipelines, enabling smooth data ingestion, transformation, and loading across multiple systems.
- Performed data transformation and preprocessing to ensure data quality, consistency, and readiness for downstream analytics and machine learning tasks.
- Built and prepared datasets for machine learning models using Spark MLlib, with minimal experience contributing to predictive analytics and model training.
- Conducted data analysis and visualization with Pandas and Matplotlib, delivering actionable insights for business and technical teams.
- Supported database operations in MySQL and PostgreSQL, ensuring efficient data retrieval and storage management.
- Collaborated with teams to improve data reliability and performance, following best practices in distributed data handling.
- **Skills:** Python, PySpark, Spark MLlib(Basics), Database management (MYSQL, PostgreSQL)

MURTI Research Center

Jan 2025 – Mar 2025

Student Research Intern

Bengaluru, India

- Researched academic papers and studies from Google Scholar and OpenAthens to build a strong foundation on the Urban Heat Island (UHI) effect and identify key environmental factors influencing urban temperatures.
- Analyzed Bengaluru's UHI effect by processing satellite imagery and geospatial datasets to map and study temperature variations across the city.
- Utilized Google Earth Engine (GEE) for efficient retrieval and preprocessing of large-scale satellite data, and applied ArcMap for spatial data analysis, mapping, and visualization.
- Developed a robust temperature prediction model using Random Forest Regression, incorporating multiple environmental factors to enhance model accuracy and reliability.
- Identified UHI hotspots and generated actionable insights to support urban planning and climate adaptation strategies, contributing to sustainable city development.
- **Skills:** Random Forest Regression

TECHNOLOGIES

Programming Languages: Python

Data Analysis and Visualization: Pandas, Matplotlib

Database Management: MySQL, PostgreSQL

Big Data and ML Tools : PySpark, Spark MLIB(Basics)

Development Tools: PyCharm, PgAdmin

CERTIFICATIONS

- Programming for Everybody (Getting Started with Python) [Certificate Link](#)
- Python for Data Science, AI and Development [Certificate Link](#)